LEAP 2025

Mathematics

Practice Test

Grade 7

DEPARTMENT of EDUCATION

Louisiana Believes

Session 1

Directions:

Today, you will take Session 1 of the Grade 7 Mathematics Practice Test. You will not be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ▶

1. Jordan's dog weighs p pounds. Emmett's dog weighs 25% more than Jordan's dog.

Which expressions represent the weight, in pounds, of Emmett's dog?

Select **each** correct answer.

A. $0.25p$

B. $1.25p$

C. $p + 0.25$

D. $p + 1.25$

E. $p + 0.25p$

GO ON ►

2. Which situation can be represented by the equation $1\frac{1}{4} \times 6 = 7\frac{1}{2}$?

A. It took Calvin $1\frac{1}{4}$ hours to run 6 miles. He ran $7\frac{1}{2}$ miles per hour.

B. Sara read for $1\frac{1}{4}$ hours every day for 6 days. She read for a total of $7\frac{1}{2}$ hours.

C. Matthew addressed $1\frac{1}{4}$ envelopes in 6 minutes. He addressed $7\frac{1}{2}$ envelopes per minute.

D. It took Beth $1\frac{1}{4}$ minutes to paint 6 feet of a board. She painted a total of $7\frac{1}{2}$ feet of the board.

GO ON ▶

3. Stefanie bought a package of pencils for $1.75 and some erasers that cost $0.25 each. She paid a total of $4.25 for these items, before tax.

Exactly how many erasers did Stefanie buy?

Enter your answer in the box.

GO ON ▶

4. The amount Troy charges to mow a lawn is proportional to the time it takes him to mow the lawn. Troy charges $30 to mow a lawn that took him 1.5 hours to mow.

Which equation models the amount in dollars, *d*, Troy charges when it takes him *h* hours to mow a lawn?

A. $d = 20h$

B. $h = 20d$

C. $d = 45h$

D. $h = 45d$

GO ON ►

5. On Monday, the temperature at 10 a.m. at Sam's house was $^{-}6°$ Fahrenheit. The temperature at 2 p.m. at Sam's house was 2° Fahrenheit. Which statement about the change in temperature from 10 a.m. to 2 p.m. at Sam's house is true?

A. The temperature decreased by 12° Fahrenheit.

B. The temperature decreased by 4° Fahrenheit.

C. The temperature increased by 3° Fahrenheit.

D. The temperature increased by 8° Fahrenheit.

GO ON ►

6. A meteorologist was monitoring the temperature outside in degrees Fahrenheit (°F) and wrote the expression $78 + (^{-}6) - 5$. Which statement best describes this expression?

A. The temperature started at 78°F and increased by 6°F. Then the temperature decreased by 5°F.

B. The temperature started at 78°F and increased by 6°F. Then the temperature increased by 5°F.

C. The temperature started at 78°F and decreased by 6°F. Then the temperature decreased by 5°F.

D. The temperature started at 78°F and decreased by 6°F. Then the temperature increased by 5°F.

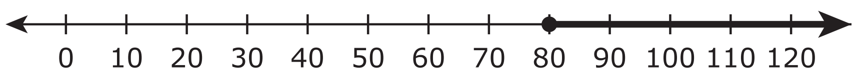
GO ON ▶

7. Ali is collecting signatures for a petition.

- He currently has 520 signatures.
- He has 6 more weeks to collect the remaining signatures he needs.
- He needs a total of at least 1,000 signatures before he can submit the petition.

Ali wants to collect the same number of signatures each week.

Which number line represents all possible numbers of signatures Ali could collect in each of the remaining weeks so that he will have enough signatures to submit the petition?

A.


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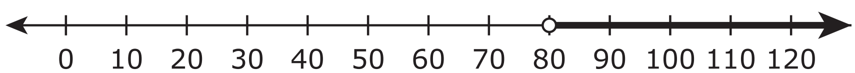

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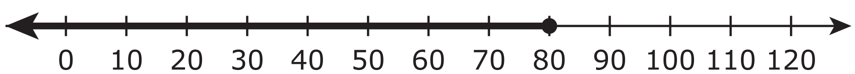

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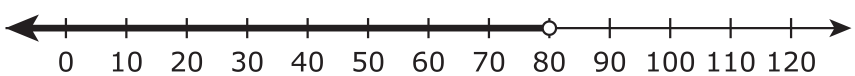


GO ON ▶

8. An ice cream shop uses a mix of blueberries and cherries on its ice cream sundaes. The shop has $5\frac{3}{4}$ pounds of blueberries and $4\frac{1}{2}$ pounds of cherries. The shop mixes the blueberries and cherries and uses $\frac{1}{16}$ pound of the mix on each sundae. Which expression represents the total number of sundaes that the shop can make using all of the blueberries and cherries?

A. $\left(5\frac{3}{4} \div \frac{1}{16}\right) + 4\frac{1}{2}$

B. $5\frac{3}{4} + \left(4\frac{1}{2} \div \frac{1}{16}\right)$

C. $\frac{1}{16} \div \left(5\frac{3}{4} + 4\frac{1}{2}\right)$

D. $\left(5\frac{3}{4} + 4\frac{1}{2}\right) \div \frac{1}{16}$

GO ON ▶

9. The table shows the cost of downloading songs from a Web site.

Cost of Songs

Number of Songs	Total Cost
3	$3.21
5	$5.35
8	$8.56

At this rate, what is the cost per song?

Enter your answer in the box.

$ [ ]

GO ON ▶

10. Which expression is equivalent to $2.2 - 2.5$?

A. $2.5 - 2.2$

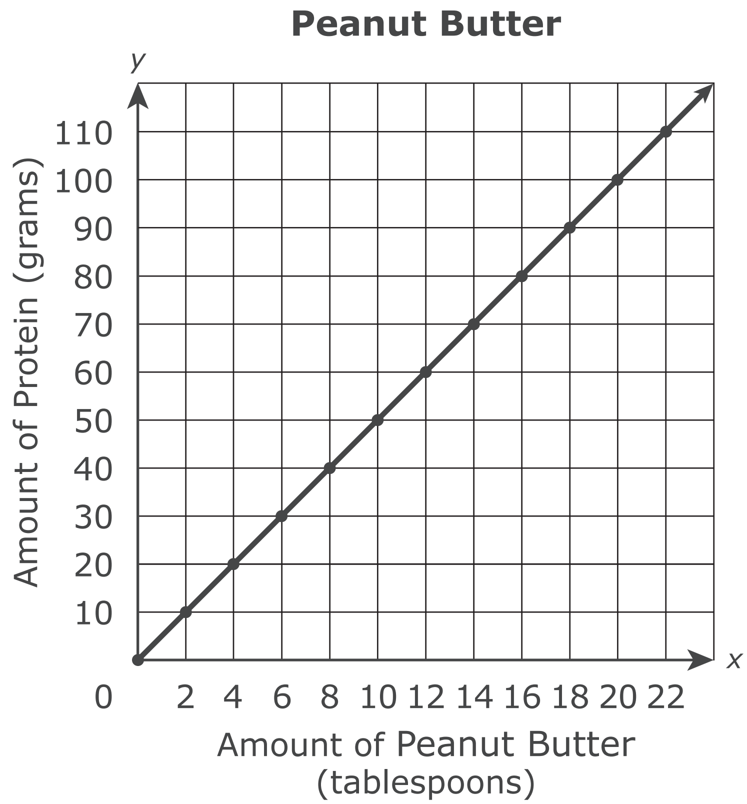
B. $2.2 + 2.5$

C. $2.2 + (^{-}2.5)$

D. $2.2 - (^{-}2.5)$

GO ON ►

11. The graph shows the amount of protein contained in a certain brand of peanut butter.



Which statement describes the meaning of the point (6, 30) on the graph?

A. There are 6 grams of protein per tablespoon of peanut butter.

B. There are 30 grams of protein per tablespoon of peanut butter.

C. There are 6 grams of protein in 30 tablespoons of peanut butter.

D. There are 30 grams of protein in 6 tablespoons of peanut butter.

GO ON ▶

12. Select the correct number from each list to complete the equation.

$$\frac{7}{8} - \left({}^{-}2 + \frac{3}{4}\right) = (\underline{\qquad\qquad} + \underline{\qquad\qquad}) + \frac{7}{8}$$

2	2
${}^{-}2$	${}^{-}2$
$\frac{3}{4}$	$\frac{4}{3}$
${}^{-}\frac{4}{3}$	${}^{-}\frac{3}{4}$

GO ON ►

13. Indicate whether each expression in the table is equivalent to $\frac{1}{2}x - 1$, equivalent to $x - \frac{1}{2}$, or **not** equivalent to $\frac{1}{2}x - 1$ or $x - \frac{1}{2}$.

Mark an X in the appropriate cells in the table. Mark one cell per row.

	Equivalent to $\frac{1}{2}x - 1$	**Equivalent to** $x - \frac{1}{2}$	**Not Equivalent to** $\frac{1}{2}x - 1$ **or** $x - \frac{1}{2}$
$\frac{2}{3}\left(\frac{3}{4}x - \frac{3}{2}\right)$			
$\left(2x + 1\right) - \left(x + \frac{3}{2}\right)$			

GO ON ►

14. Two equations are shown.

Equation 1: $\frac{2}{3}\left(x - 6\right) = 6$

Equation 2: $\frac{2}{3}y - 6 = 6$

Solve each equation. Then, enter a number in each box to make this statement true.

The value of x is $\boxed{}$, and the value of y is $\boxed{}$.

GO ON ▶

15. Jessica rented 1 video game and 3 movies for a total of $11.50.

- The video game cost $4.75 to rent.
- The movies cost the same amount each to rent.

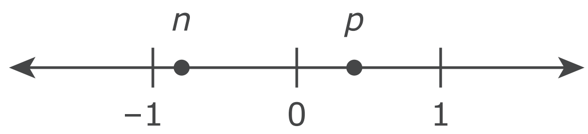
What amount, in dollars, did Jessica pay to rent each movie?

Enter your answer in the box.

$ | | |
|---|
| |

GO ON ▶

16. Two numbers, n and p are plotted on the number line shown.



The numbers $n - p$, $n + p$, and $p - n$ will be plotted on the number line.

Select an expression from each list to make this statement true.

The number with the least value is ________________,

$n - p$
$n + p$
$p - n$

and the number with the greatest value is ________________.

$n - p$
$n + p$
$p - n$

GO ON ▶

17. An airplane's altitude changed $^{-}378$ feet over 7 minutes. What was the mean change of altitude in feet per minute?

Enter your answer in the box.

GO ON ▶

18. Anita earns 60 points every time she shops at a grocery store. She needs a total of 2,580 points to receive a free prize. So far, she has earned 480 points. How many more times will Anita have to shop at the grocery store in order to earn the additional points she needs for a free prize?

A. 8

B. 35

C. 43

D. 51

GO ON ▶

19. The table below represents a relationship between the time a turtle walks and the distance the turtle travels.

Time and Distance Turtle Walks

Time (minutes)	Distance (feet)
5	120
20	480
30	720
50	1,200

What is the unit rate, in feet per minute, represented in this table?

Enter your answer in the box.

GO ON ▶

20. Which expressions are equivalent to the expression $(x - y)\frac{5}{8} - \frac{1}{4}x + y$?

Select **each** correct answer.

A. $\frac{3}{8}x + \frac{3}{8}y$

B. $\frac{3}{8}x + 1\frac{5}{8}y$

C. $\frac{5}{8}x - y - \frac{1}{4}x + y$

D. $\frac{5}{8}x - \frac{5}{8}y - \frac{1}{4}x + y$

E. $\frac{5}{8}x - \frac{1}{4}x + y - \frac{5}{8}y$

STOP

Session 2

Directions:

Today, you will take Session 2 of the Grade 7 Mathematics Practice Test. You will be able to use a calculator in this session.

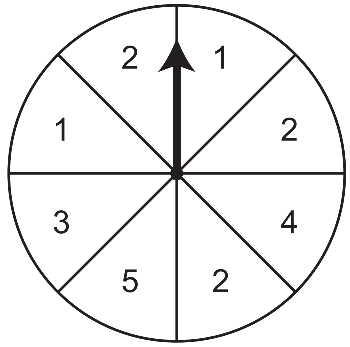
Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ▶

21. The spinner shown is divided into 8 equal sections.



The arrow on this spinner is spun once.

What is the probability that the arrow will land on a section labeled with a number **greater** than 3?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

GO ON ▶

22. Jamal will slice a right circular cylinder into two congruent pieces. Which two-dimensional plane sections **could result** from the slice Jamal makes?

Select **each** correct answer.

A. circle

B. pentagon

C. hexagon

D. triangle

E. rectangle

GO ON ▶

23. Rosy waxes $\frac{2}{3}$ of her car with $\frac{1}{4}$ bottle of car wax.

At this rate, what fraction of the bottle of car wax will Rosy use to wax her entire car?

A. $\frac{1}{8}$

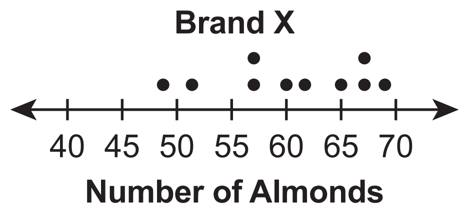
B. $\frac{1}{6}$

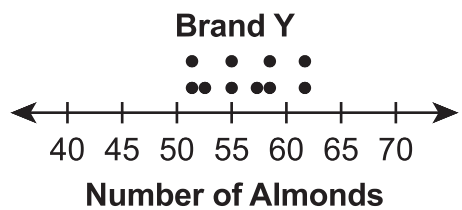
C. $\frac{3}{8}$

D. $\frac{3}{4}$

GO ON ▶

24. Alexis chose a random sample of 10 jars of almonds from each of two different brands, X and Y. Each jar in the sample was the same size. She counted the number of almonds in each jar. Her results are shown in the plots.





Based on the plots, which statement **best** compares the numbers of almonds in the jars from the two brands?

A. The number of almonds in jars from Brand X tends to be greater and more consistent than those from Brand Y.

B. The number of almonds in jars from Brand X tends to be greater and less consistent than those from Brand Y.

C. The number of almonds in jars from Brand X tends to be fewer and more consistent than those from Brand Y.

D. The number of almonds in jars from Brand X tends to be fewer and less consistent than those from Brand Y.

GO ON ►

25. A right triangle has legs measuring 4.5 meters and 1.5 meters.

The lengths of the legs of a second triangle are proportional to the lengths of the legs of the first triangle.

Which could be the lengths of the legs of the second triangle?

Select **each** correct pair of lengths.

A. 6 m and 2 m

B. 8 m and 5 m

C. 7 m and 3.5 m

D. 10 m and 2.5 m

E. 11.25 m and 3.75 m

GO ON ►

26. A $4\frac{1}{2}$-ounce hamburger patty has $25\frac{1}{2}$ grams of protein, and 6 ounces of fish has 32 grams of protein. Determine the grams of protein per ounce for each type of food.

Select a number from each list to correctly complete each statement.

A hamburger patty has approximately ________________ grams of protein per ounce.

0.2
4.5
5.7
21.0
25.5

The fish has approximately ________________ grams of protein per ounce.

0.2
5.3
6.0
26.0
32.0

GO ON ▶

27. A salesperson earns commission on the sales that she makes each month.

- The salesperson earns a 5% commission on the first $5,000 she has in sales.
- The salesperson earns a 7.5% commission on the amount of her sales that are greater than $5,000.

Part A

This month the salesperson had $8,000 in sales. What amount of commission, in dollars, did she earn?

A. $400

B. $475

C. $525

D. $600

Part B

The salesperson earned $1,375 in commission last month. How much money, in dollars, did she have in sales last month?

Enter your answer in the box.

GO ON ►

28. The attendance for the last 4 years at a county fair is shown in the table.

County Fair Attendance

Year	Attendance
1	9,278
2	10,365
3	12,128
4	13,304

This year, the first 20% of people attending the fair will receive a raffle ticket. Of the people who receive raffle tickets, $\frac{1}{3}$ will receive a small prize.

- Based on the data in the table, determine a reasonable estimate of the number of people who will attend this year's fair. Explain how you found your estimate.
- Use your estimate to find the approximate number of people who will receive a small prize at this year's fair.
- Show your work or provide an explanation of how you found the approximate number of people who will receive a small prize at this year's fair.

Enter your answers and your work or explanations in the box provided.

GO ON ▶

29. A family purchased tickets to a museum and spent a total of $38.00. The family purchased 4 tickets. There was a $1.50 processing fee for each ticket. Write and solve an equation that can be used to find x, the cost of one ticket to the museum. Show your work or explain your answer.

Enter your equation, your answer, and your work or explanation in the box provided.

GO ON ►

30. Consider the equation $5 + x = n$.

What must be true about any value of x if n is a negative number? Explain your answer. Include an example with numbers to support your explanation.

Enter your answer, your explanation, and your example in the box provided.

GO ON ▶

31. Part A

Which sets of measurements could be the interior angle measures of a triangle?

Select **each** correct answer.

A. 10°, 10°, 160°

B. 15°, 75°, 90°

C. 20°, 80°, 100°

D. 35°, 35°, 105°

E. 60°, 60°, 60°

Part B

Which sets of measurements could be the side lengths of a triangle?

Select **each** correct answer.

A. 3 cm, 3 cm, 3 cm

B. 4 cm, 8 cm, 13 cm

C. 5 cm, 9 cm, 14 cm

D. 6 cm, 7 cm, 8 cm

E. 7 cm, 7 cm, 10 cm

GO ON ►

32. The coordinates of a quadrilateral are shown:

- point J ($^{-}4.5$, 3)
- point K ($^{-}1.2$, 3)
- point L ($^{-}1.2$, 8.7)
- point M ($^{-}4.5$, 8.7)

Brenda claims that quadrilateral *JKLM* is a square.

Part A

Show or explain why Brenda is not correct.

Enter your work or explanation in the box provided.

GO ON ▶

Part B

Select new coordinates for point *L* and point *M* so that quadrilateral *JKLM* is a square. Show or explain all of the steps you used to determine the new locations of the two points.

Enter your answers and your work or explanation in the box provided.

STOP

Session 3

Directions:

Today, you will take Session 3 of the Grade 7 Mathematics Practice Test. You will be able to use a calculator in this session.

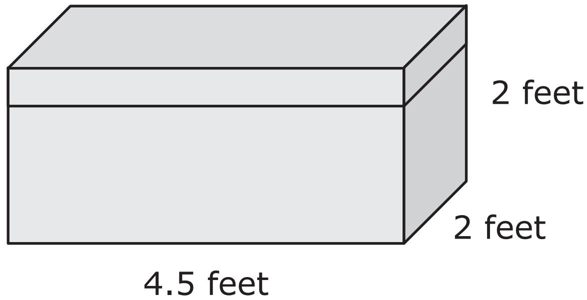
Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ►

33. A storage chest is shown.



What are the volume and the surface area of this storage chest?

Enter your answers in the boxes.

Volume = [ ] cubic feet

Surface Area = [ ] square feet

GO ON ►

34. Josephine owns a diner that is open every day for breakfast, lunch, and dinner. She offers a regular menu and a menu with daily specials. She wanted to estimate the percentage of her customers who order specials. She selected a random sample of 50 customers who had lunch at her diner during a three-month period. She determined that 28% of these customers ordered from the menu with specials.

Which statement about Josephine's sample is true?

A. The sample is the percentage of customers who ordered daily specials.

B. The sample might not be representative of the population because it only included lunch customers.

C. The sample shows that exactly 28% of Josephine's customers ordered daily specials.

D. No generalizations can be made from this sample, because the sample size of 50 is too small.

GO ON ►

35. The table shows a proportional relationship between the number of pounds of grapes purchased and the total cost of the grapes.

Grapes

Number of Pounds	Total Cost (dollars)
4	2.76
7	4.83
9	6.21

A row of values is missing in the table.

Which number of pounds of grapes and total cost of the grapes could be used as the missing values in the table?

Select **each** correct response.

A. Pounds of grapes: 2
Total cost: $1.38

B. Pounds of grapes: 3
Total cost: $2.53

C. Pounds of grapes: 6
Total cost: $3.68

D. Pounds of grapes: 8
Total cost: $5.52

E. Pounds of grapes: 11
Total cost: $8.97

GO ON ▶

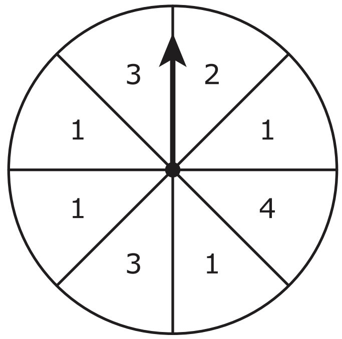
36. A student usually saves \$20 a month. He would like to reach a goal of saving \$350 in 12 months. The student writes the equation $350 = 12(x + 20)$ to represent this situation. Solve the equation for x.

- Show your work or explain your answer.
- Write your answer as a sentence that describes what the variable x represents.

Enter your answers and your work or explanation in the box provided.

GO ON ▶

37. This spinner is divided into eight equal-sized sections. Each section is labeled with a number.



Jake spins the arrow on the spinner once.

Write events in the correct order from least likely to most likely.

Events

Arrow lands on a section labeled with an odd number

Arrow lands on a section labeled with the number 1.

Arrow lands on a section labeled with a number less than 4.

Least Likely

Most Likely

GO ON ▶

38. Jonah has a recipe that uses $1\frac{1}{2}$ cups of brown sugar and $2\frac{1}{3}$ cups of flour to make 24 muffins. He has a total of 7 cups of flour. Jonah wants to use all of his flour to make as many muffins as possible using this recipe.

Part A

Exactly how many cups of brown sugar will Jonah use if he uses all 7 cups of flour?

A. $3\frac{3}{10}$ cups

B. $4\frac{1}{2}$ cups

C. $7\frac{5}{6}$ cups

D. $10\frac{8}{9}$ cups

Part B

Exacly how many muffins will Jonah make if he uses all 7 cups of flour?

Enter your answer in the box.

GO ON ▶

39. Reagan will use a random number generator 1,200 times. Each result will be a digit from 1 to 6. Which statement **best** predicts how many times the digit 5 will appear among the 1,200 results?

A. It will appear exactly 200 times.

B. It will appear close to 200 times but probably not exactly 200 times.

C. It will appear exactly 240 times.

D. It will appear close to 240 times but probably not exactly 240 times.

GO ON ►

40. Part A

At Fairview Middle School, 75 band members need to raise a total of $8,250 for a trip. So far, they have raised $3,120.

How much money, in dollars, per band member, still needs to be raised for the trip?

Enter your answer in the box.

Part B

The entire band decides to have a concert to raise the money for the trip. Tickets for the concert will cost $7.50 each. A local business agrees to donate an additional $0.50 for each $1.00 in ticket sales to the band for their trip.

What is the **least** number of concert tickets the band must sell in order to raise the rest of the money needed for the trip?

Enter your answer in the box.

GO ON ▶

41. Misha has a cube and a right-square pyramid that are made of clay. She placed both clay figures on a flat surface.

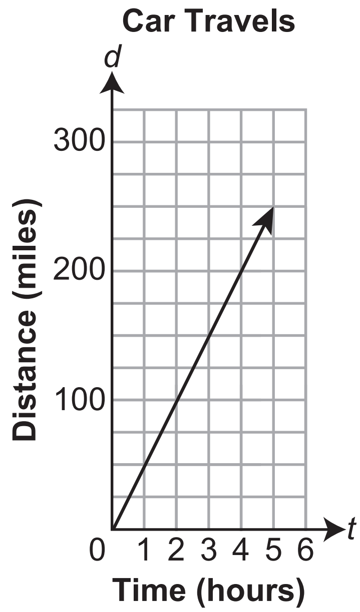
Mark an X in **each** box in the table that identifies the two-dimensional-plane sections that **could** result from a vertical or horizontal slice through each clay figure.

	Cube	**Right-Square Pyramid**
Triangle		
Square		
Rectangle That is Not a Square		

GO ON ▶

42. Part A

The graph shows the distance in miles, d, a car travels in t hours.



Explain why the graph does or does not represent a proportional relationship between the variables d and t.

Enter your explanation in the box provided.

GO ON ▶

Part B

Two cars leave from the same city at the same time and drive in the same direction. The table shows the distances traveled by each car.

Two Cars Travel

Hours of Travel	Miles Traveled by Red Car	Miles Traveled by White Car
1	77	55
2	122	110
3	167	165
4	212	220
5	257	275

- Determine whether the relationship between the number of hours traveled and the number of miles traveled is proportional for each car.
- Use the table to explain how you determined your answers.
- Describe how the graph of the distance traveled by each car would support your answers.

GO ON ►

Enter your answers and your explanations in the box provided.

GO ON ►

43. A worker has to drive her car as part of her job. She receives money from her company to pay for the gas she uses. The table shows a proportional relationship between *y*, the amount of money that the worker receives, and *x*, the number of work-related miles driven.

Mileage Rates

Distance Driven, *x* (miles)	Amount of Money Received, *y* (dollars)
25	12.75
35	17.85
40	20.40
50	25.50

GO ON ►

Part A

Explain how to compute the amount of money the worker receives for any number of work-related miles. Based on your explanation, write an equation that can be used to determine the total amount of money, y, the worker receives for driving x work-related miles.

Enter your explanation and your equation in the box provided.

GO ON ▶

Part B

On Monday, the worker drove a total of 134 work-related and personal miles. She received $32.13 for the work-related miles she drove on Monday. What percent of her total miles driven were work-related on Monday? Show or explain your work.

Enter your answer and your work or explanation in the box provided.

STOP

DEPARTMENT of EDUCATION
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LEAP 2025 Grade 7 Mathematics Practice Test Answer Key

LEAP 2025

This document contains the answer keys and rubrics for the LEAP 2025 Grade 7 Mathematics Practice Test.

Session 1				
Task #	**Task Type**	**Value (points)**	**Key**	**Alignment**
1	I	1	B, E	7.EE.A.2
2	I	1	B	7.NS.A.2a
3	I	1	10	7.EE.B.4a
4	I	1	A	7.RP.A.2c
5	I	1	D	7.NS.A.3
6	I	1	C	7.NS.A.1b
7	I	1	A	7.EE.B.4b
8	I	1	D	7.NS.A.2b
9	I	1	1.07	7.RP.A.2b
10	I	1	C	7.NS.A.1c
11	I	1	D	7.RP.A.2d
12	I	1	$^{7}/_{8} - (-2 + {}^{3}/_{4}) = ($ 2 $+$ –3/4 $) + {}^{7}/_{8}$	7.NS.A.1d
13	I	1	Columns: Equivalent to $\frac{1}{2}x-1$ / Equivalent to $x-\frac{1}{2}$ / Not Equivalent to $\frac{1}{2}x-1$ or $x-\frac{1}{2}$ $\frac{2}{3}\left(\frac{3}{4}x-\frac{3}{2}\right)$: ✔ / ☐ / ☐ $(2x+1)-\left(x+\frac{3}{2}\right)$: ☐ / ✔ / ☐	7.EE.A.1
14	I	1	15; 18	7.EE.B.4a
15	I	1	2.25	7.EE.B.4a
16	I	1	The number with the least value is n – p, and the number with the greatest value is p – n.	7.NS.A.1b
17	I	1	-54	7.NS.A.3
18	I	1	B	7.EE.B.4a
19	I	1	24	7.RP.A.2b
20	I	1	A, D, E	7.EE.A.1

Session 2				
Task #	**Task Type**	**Value (points)**	**Key**	**Alignment**
21	I	1	B	7.SP.C.7a
22	I	1	A, E	7.G.A.3
23	I	1	C	7.RP.A.1
24	I	1	B	7.SP.B.4
25	I	1	A, E	7.RP.A.2a
26	I	1	A hamburger patty has approximately 5.7 grams of protein per ounce. The fish has approximately 5.3 grams of protein per ounce.	7.RP.A.1
27	I	2	Part A: B Part B: 20000	7.RP.A.3
28	III	3	rubric	LEAP.III.7.4 (7.NS.A.3, 7.EE.B.3)
29	III	3	rubric	LEAP.III.7.1 (7.EE.B.4a)
30	II	3	rubric	LEAP.II.7.2 (7.NS.A.1b)
31	I	2	Part A: A, B, E Part B: A, D, E	7.G.A.2
32	II	4	Part A: rubric Part B: rubric	LEAP.II.7.6 (6.NS.C.6b, 6.NS.C.8)

Session 3				
Task #	**Task Type**	**Value (points)**	**Key**	**Alignment**
33	I	1	18; 44	7.G.B.6
34	I	1	B	7.SP.A.1
35	I	1	A, D	7.RP.A.2a
36	II	3	rubric	LEAP.II.7.5 (7.EE.B.4a)

<table>
<tr><th colspan="5">Session 3</th></tr>
<tr><th>Task #</th><th>Task Type</th><th>Value (points)</th><th>Key</th><th>Alignment</th></tr>
<tr><td>37</td><td>I</td><td>1</td><td>Least Likely
Arrow lands on a section labeled with the number 1.
Arrow lands on a section labeled with an odd number.
Arrow lands on a section labeled with a number less than 4.
Most Likely

Note: This item presents horizontally online.</td><td>7.SP.C.5</td></tr>
<tr><td>38</td><td>I</td><td>2</td><td>Part A: B
Part B: 72</td><td>7.RP.A.3</td></tr>
<tr><td>39</td><td>I</td><td>1</td><td>B</td><td>7.SP.C.6</td></tr>
<tr><td>40</td><td>I</td><td>2</td><td>Part A: 68.40
Part B: 456</td><td>7.EE.B.3</td></tr>
<tr><td>41</td><td>I</td><td>1</td><td><table><tr><th></th><th>Cube</th><th>Right-Square Pyramid</th></tr><tr><td>Triangle</td><td>☐</td><td>☑</td></tr><tr><td>Square</td><td>☑</td><td>☑</td></tr><tr><td>Rectangle That Is Not a Square</td><td>☑</td><td>☐</td></tr></table></td><td>7.G.A.3</td></tr>
<tr><td>42</td><td>II</td><td>4</td><td>Part A: rubric
Part B: rubric</td><td>LEAP.II.7.4 (7.RP.A.2a)</td></tr>
<tr><td>43</td><td>III</td><td>6</td><td>Part A: rubric
Part B: rubric</td><td>LEAP.III.7.2 (6.RP.A.2, 6.RP.A.3, 6.EE.C.9)</td></tr>
</table>

RUBRICS

Task # 28	
Score	**Description**

3	Student response includes the following 3 elements:
	• **Computation component**: 1 point
	    ○ Acceptable approximate number of people who will receive a small prize, range from 900 to 1,200 people
	• **Modeling component**: 2 points
	    ○ Models a valid estimation strategy for determining the number of people who will attend this year's fair, range of 14,000 to 17,000
	    ○ Models finding the approximate number of people who will receive a prize
	
	Sample Student Response:
	I saw that the attendance was increasing each year and found the average amount that it increased by each year.
	(1,087 + 1,763 + 1,176)/3 = 4,026/3
	So I estimate that the attendance this year will increase by about 1,342 people and will be 14,646 people.
	
	20% of 14,646 is 0.20(14,468) = 2,929.2
	
	$\frac{1}{3}$ of 2,929.2 is (2,929.2)($\frac{1}{3}$) = (2,929.2)/3 = 976.4
	
	So about 976 people will receive a small prize.
	
	Note: Accept other valid estimation strategies for determining this year's attendance.

2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

<table>
<tr><th colspan="2">Task #29</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
Modeling component: 2 points
Correct equation
Valid explanation or work
Computation component: 1 point
Correct price of one museum ticket, 8
Sample Student Response:
$4(x + 1.50) = 38$ or equivalent
$4x + 6 = 38$
$4x = 32$
$x = 8$

The cost of one ticket is $8.</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #30</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
Reasoning component: 2 points
Valid statement about the value of x
Valid explanation about the statement regarding the value of x
Computation component: 1 point
Valid example, using numbers, that supports the explanation
Sample Student Response:
I know that $5 + (-5) = 0$. Then, 5 plus any number less than -5 will be negative. So, the value of x must be less than -5 if n is a negative number ($x < -5$ can be used as the statement). An example that shows this is true is $5 + (-6) = -1$, and this works for any number less than -5.</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #32</th></tr>
<tr><th colspan="2">Part A</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>2</td><td>Student response includes the following 2 elements:

Computation component: 1 point
Correct computation, numerical support, or graphical support that is consistent with the student’s reasoning
Reasoning component: 1 point
Correctly reasons that the lengths of the sides of the quadrilateral JKLM are not all the same, so it cannot be a square

Sample Student Response:
In a square, the lengths of all four sides are the same. If quadrilateral JKLM is a square, all four of its side lengths would be the same. Since the y-coordinates are the same in points J and K, the side length of JK is the positive difference between the x-coordinates of each point. So, JK = $|-4.5-(-1.2)| = |-4.5+1.2| = |-3.3| = $ 3.3 units. Similarly, the side length of KL is the positive difference between the y-coordinates of each point. So, KL = $|3-8.7| = |-5.7|$ = 5.7 units. The lengths of two sides of the quadrilateral are not equal, so quadrilateral JKLM is not a square.

Notes:

The student may still receive credit for this part if the student chooses to compute or compare side lengths without using absolute values.
The student may receive a total of 1 point for Part A if the reasoning processes are correct but the student makes one or more computational errors resulting in incorrect answers or an incorrect conclusion.
The student may receive the 1 computation point if the correct answer is computed but shows no work or insufficient work to indicate a correct reasoning process.
</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 2 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #32</th></tr>
<tr><th colspan="2">Part B</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>2</td><td>Student response includes the following 2 elements:
• Computation component: 1 point
  ○ Correct new coordinates for points L and M
• Reasoning component: 1 point
  ○ Correctly reasons why the two new coordinates of points L and M would make quadrilateral JKLM a square
  Note: Numerical or graphical support that is consistent with the student’s reasoning is acceptable for full credit.

Sample Student Response:
The given coordinates form a rectangle with sides JK and LM both 3.3 units and sides KL and JM both 5.7 units. If the coordinates of points L and M change so that quadrilateral JKLM is a square, they should be lowered on the coordinate plane 5.7 – 3.3, or 2.4 units. This will change sides KL and JM from 5.7 units to 3.3 units, making the resulting quadrilateral a square. Lowering points on a coordinate plane changes their y-coordinates. So, the new coordinates of point L would be (-1.2, 6.3) since 8.7 – 2.4, or 6.3. The new coordinates of point M would be (-4.5, 6.3) since 8.7 – 2.4, or 6.3 units.

Notes:
• The student should receive credit for this part if the student chooses new coordinates for points L and M that are below points J and K, as long as the student shows or explains that the side lengths of all four sides are the same length.
• The student may receive a total of 1 point for Part B if the reasoning processes are correct but the student makes one or more computational errors resulting in incorrect answers or an incorrect conclusion.
• The student may receive the 1 computation point if the correct answer is computed but shows no work or insufficient work to indicate a correct reasoning process.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 2 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #36</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:

Computation component: 1 point
Correctly determines the value of x
Reasoning component: 2 points
Correctly uses an equation to determine the monthly savings goal
Correctly writes a sentence to explain the solution

Sample Student Response:
$350 = 12(x + 20)$
$29.1\overline{6} = x + 20$
$9.1\overline{6} = x$
$\$9.17 = x$

The student has to save an additional $9.17 per month to reach his goal of saving $350 in 12 months.</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #42</th></tr>
<tr><th colspan="2">Part A</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>1</td><td>Student response includes the following element:
Reasoning component: 1 point
Correct explanation of why the graph represents a proportional relationship
Sample Student Response:
The graph represents a proportional relationship between the variables d and t because the ratio of d to t is always the same number.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
<tr><th colspan="2">Part B</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
Computation component: 1 point
Correct identification of the relationship of distance and time as proportional for the white car and not proportional for the red car
Reasoning component: 2 points
Correct explanation, using the table, of why each relationship is proportional or not proportional
Correct explanation of how the graph of each relationship would support the previous answer
Sample Student Response:
The relationship between distance and time is proportional for the white car, but not proportional for the red car. The ratio of miles traveled to hours traveled for the white car is the same for each row (55 miles per hour). The ratio of miles traveled to hours traveled for the red car is not the same for each row ($\frac{77}{1}$ = 77, and $\frac{122}{2}$ = 61). The graph of the white car relationship would form a straight line that passes through the origin, so this supports my answer that it is a proportional relationship. The graph of the red car relationship would also pass through the origin, but does not form a straight line. This also supports my answer that the red car relationship is not a proportional relationship.</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>

<table>
<tr><th colspan="2">Task #43</th></tr>
<tr><th colspan="2">Part A</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
• Computation component: 1 points
    ○ Correct amount of money received for each work-related mile driven, $0.51
• Modeling component: 2 points
    ○ Explanation of how to find the amount of money received for any number of work-related miles driven
    ○ Correct equation based on the explanation given

Sample Student Response:
Since the table shows a proportional relationship, I can divide the amount of money received by the distance driven for any of the rows in the table. The worker received $0.51 for each work-related mile driven. The equation that represents this is $y = 0.51x$ (or equivalent).</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
<tr><th colspan="2">Part B</th></tr>
<tr><th>Score</th><th>Description</th></tr>
<tr><td>3</td><td>Student response includes the following 3 elements:
• Computation component: 2 points
    ○ Correct number of work-related miles driven, 63
    ○ Correct percent of total miles driven: 47% (or correct calculation based on incorrect number of work-related miles driven)
• Modeling component: 1 point
    ○ Correct explanation given or work shown

Sample Student Response:
The percent of total miles is found by dividing the work-related miles driven by the total number of miles driven. So, I must first determine the total number of miles that were work-related. I can use my equation from Part A to find the answer.
$32.13 = 0.51x$
$x = \frac{32.13}{0.51} = 63$
$\frac{63}{134} \times 100 \approx 47\%$</td></tr>
<tr><td>2</td><td>Student response includes 2 of the 3 elements.</td></tr>
<tr><td>1</td><td>Student response includes 1 of the 3 elements.</td></tr>
<tr><td>0</td><td>Student response is incorrect or irrelevant.</td></tr>
</table>